

Liam Meagher

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EDUCATION

University of Southern California, M.S. in Mechanical Engineering **Expected May 2028**

University of California, Los Angeles, B.S. in Computer Science & Engineering **September 2021 – June 2025**

EXPERIENCE

Chassis Lead Engineer, Flagship Combat Robotics – Los Angeles, CA **January 2023 – June 2025**

- Led the mechanical design and fabrication of **4** autonomous vertical and horizontal spinner weapon systems
- Applied GD&T and tolerance stack-up analysis to achieve precise fits, alignment, and **20+** DFM-ready component designs
- Used SolidWorks CAD and FEA to validate the design integrity of chassis subassemblies under high-shock loading (**11+ kN**)
- Employed Ansys Discovery topological optimization to reduce mass by **42% (10.3 lb)** while maintaining stiffness targets
- Created over **50** fabrication drawings and release packages with datums, tolerances, and inspection-critical dimensions
- Manufactured critical **AR500** steel and **7075-T6** aluminum components utilizing both manual and CNC milling operations
- Accelerated chassis and drivetrain prototyping cycles by **30%** through rapid iteration with 3D printing and laser cutting
- Instructed 30+ team members on researching, designing, and machining **4** competition-ready combat robots

Autonomous Systems Software Developer, Bruin Underwater Robotics – Los Angeles, CA **September 2023 – June 2024**

- Developed autonomy control software for an underwater robot, implementing real-time feedback control loops for depth
- Improved closed-loop stability by reducing oscillations by **40%** through PID tuning and dynamic response testing
- Generated **2000+** simulated sensor frames and produced **5+** hours of synthetic mission data using Unity's simulation tools
- Conducted simulation-driven reliability testing across **8** mission scenarios, increasing autonomous control stability by **45%**
- Integrated control software with thrusters, depth sensors, and embedded hardware to enhance performance and reliability during autonomous underwater pathing missions using technologies such as ROS and C++

Programming & Game Design Lead, ACM Studio – Los Angeles, CA **September 2021 – January 2024**

- Developed and implemented rotating classes, structures, and colliders for **4** weapon systems and **16** unique animations
- Led rapid prototyping trials using Unity for asset integration and simulation-based debugging within strict deadlines
- Implemented AI-driven decision systems with **4** scaling difficulty levels, modeling adaptive opponent behavior and skills
- Optimized Unity's performance and reduced memory usage by **28%** to support large-scale simulation testing of opponents

Full-Stack Web Developer, UCLA Marketplace – Los Angeles, CA **January 2022 – January 2023**

- Designed and developed a peer-to-peer platform enabling **200+** UCLA students to buy, sell, and trade items without fees
- Implemented secure SQL-based authentication and transaction infrastructure, ensuring reliable data integrity across users
- Maintained shared codebase for **6** developers by implementing structured Git workflows, reducing merge conflicts by **30%**
- Achieved **98%** uptime with Node.js landing page and secure peer-to-peer messaging board across student accounts

Activities & Outdoor Programming Director, Scouting America – Avalon, CA **May 2021 – August 2022**

- Operated a weekly outdoor activity program supporting over **4,000** participants with a budget of over **\$390,000**
- Led and evaluated **30** staff members, provided constructive feedback, enhancing collaboration and communication skills
- Managed inventory for **170+** full-time employees by implementing a new tracking system, reducing supply errors by **15%**
- Created a financial report analyzing performance metrics across **10** projects, facilitating improved resource management
- Recruited and trained **30** summer staff members, improving operational efficiency and customer satisfaction ratings

SKILLS

Software: SolidWorks (CAD/FEA), Onshape (CAD/PDM), Ansys Discovery, MATLAB, Linux, Excel, Unity, Unreal Engine

Manufacturing: CNC (Mill/Lathe), 3D Printing (FDM), G-Code, Soldering, Drill Press, GD&T, Design for Manufacturability (DFM)

Languages: C, C++, C#, Haskell, Java, JavaScript, Python, R, RISC-V Assembly, SQL, Verilog

Libraries & Tools: NumPy, Pandas, Matplotlib, OpenCV, Scikit-Learn, PyTorch, React, Node.js, Docker, Git, Arduino, ROS, ROS 2